iii. Iodine number:

Iodine number is expressed as the number of grams of iodine absorbed by 100 g of fat. Not only iodine, but other halogens will also bind at double bonds. Hence bromine is often used instead of iodine because it is more reactive. The iodine number is a measure of the degree of unsaturation of the fatty acid in the fat.

Eg: the iodine numbers of linseed oil (PUFA), olive oil (MUFA), and coconut oil (saturated fatty acid) are approximately 175–201, 77–91, and 8–9.5, respectively.

iv. Polenske number:

The Polenske value indicates the level of volatile fatty acid in the fat. It is equal to the number of milliliters of 0.1N alkali (KOH) solution necessary to neutralize the water insoluble fatty acids. The value is named after the chemist who developed it, Eduard Polenske.

v. Riechert Meissl number

The Reichert value (Reichert-Meissl-Wollny number) measures the quantity of short chain fatty acid in the fat molecule. It is the number of milliliters of 0.1 N KOH solution necessary to neutralize the soluble, volatile fatty acids derived from 5 g of fat. The value is named after the chemists who developed it, Emil Reichert and EmerichMeissl.

vi. Acetyl number

Acetyl number is the measure of the free hydroxyl groups in a substance (as a fat or oil). It is the number of milligrams of potassium hydroxide required for neutralization of the acetic acid formed by the saponification of 1gm of fat after it has been acetylated.

6.4. Phospholipids

Phospholipids are compound lipids. They are amphipathic molecules, meaning they have a hydrophobic tail and a hydrophilic head group. Phospholipids are major components of the plasma membrane, which serves as a barrier between the cell and its surroundings. In a membrane, phospholipids are arranged in a structure called a lipid bilayer, with their phosphate heads facing the water and their tails pointing towards the interior of the membrane.

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